

Four Tools for the Glass-Box Toolkit: LIME, SHAP, Saliency Methods, and Counterfactual Explanations

Session 3 Reading — Artificial Cognition and Explainable AI (5-Day Intensive)

Abstract

Sessions 1 and 2 mapped the landscape of explainable AI: why opacity matters, what kinds of explanations are possible, and what regulations demand them. This reading turns from the map to the toolkit. The four techniques covered here — LIME, SHAP, visual saliency methods (Grad-CAM, SmoothGrad, Integrated Gradients), and counterfactual explanations — are the ones most likely to appear in a vendor demonstration, an audit report, or a regulatory submission. Each technique is explained in plain English, with its underlying intuition, its outputs, the audiences it serves, and its limitations. A worked example shows how the four can be applied jointly to a single clinical decision. A decision framework helps the reader choose among them on the basis of audience, data type, and the question being asked. The reading closes with a checklist of validation pitfalls every consumer of XAI outputs should keep in mind. The aim is not to teach the algorithms but to ensure the reader can read their outputs, question their claims, and judge whether a particular technique answers the question their organisation is asking.

Keywords: LIME; SHAP; saliency maps; counterfactual explanations; Shapley values

1. From principle to practice

Sessions 1 and 2 treated explainability conceptually: scope, source, motivation, taxonomy. This reading walks across that landscape and visits the techniques themselves. The four covered here — LIME, SHAP, saliency methods (Grad-CAM and its successors), and counterfactual explanations — dominate industry XAI tooling today. A loan officer, a compliance auditor, a hospital safety committee, or a journalist investigating an algorithmic harm will encounter the outputs of at least one of these techniques. Knowing what each technique does, what it does not do, and which audiences it serves is essential reading for anyone evaluating a deployment.

A note on what we leave out. The four techniques covered here all sit in the post-hoc, local quadrant of the 2×2 from Session 2. There are reasons for this concentration. Local-post-hoc tools attract the bulk of XAI research and commercial tooling because that is where regulatory and social demand is greatest — every credit denial, medical diagnosis, or hiring rejection raises a local question, often from a person seeking recourse. Intrinsic-interpretable methods (covered in Session 2: scorecards, generalised additive models, short decision trees) belong in your toolkit too, but they are model-design choices rather than post-hoc techniques to be applied to an

existing system. Cynthia Rudin's argument from Session 2 deserves a final word: the best explanation is often a model that did not need one because its structure already explained itself.¹

We will treat each technique with a running example: Maria, a loan applicant whose application has been denied by the bank's machine-learning model, and a chest X-ray that an FDA-cleared AI system has flagged as suspicious for pneumonia. The aim is not to teach you to run the algorithms yourself but to ensure you can read their outputs, question their claims, and judge whether a particular technique answers the question your organisation is asking.

2. LIME — clearing a small space in the forest

Local Interpretable Model-Agnostic Explanations (LIME), introduced by Marco Tulio Ribeiro and colleagues in 2016, was the first XAI technique to capture wide industry attention.² Its core insight is simple enough to state in one sentence: even when a model's global behaviour is hopelessly complex, the model's behaviour in a small neighbourhood around any single prediction can usually be approximated by a simple, interpretable model.

Imagine the black-box model's input–output landscape as a dense, tangled forest. You cannot survey the whole forest from any single vantage point. But around the particular tree you are interested in — the tree representing Maria's denied loan application — you can clear a small space, plant a few stakes, and survey what is around you. That *local clearing* is what LIME provides.

How LIME works, in plain English

Take the data point you want to explain: Maria's application, scored 0.31 by the bank's model and therefore denied. The LIME procedure has six steps.

1. Generate *perturbations* of Maria's application: many slightly modified versions of her record, with income nudged up or down, credit score altered, recent inquiry counts varied, and so on.
2. Score each perturbed record using the original black-box model. The model has not changed; LIME only queries it.
3. *Weight* the perturbations: those closer to Maria's actual application count more.
4. Fit a simple *interpretable model* — typically a linear regression with a small number of non-zero coefficients — to predict the black-box scores from the perturbed features, with the weights from Step 3.
5. Report the *coefficients* of the simple model as the explanation.

¹Cynthia Rudin, 'Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead' (2019) 1(5) *Nature Machine Intelligence* 206.

²Marco Tulio Ribeiro, Sameer Singh and Carlos Guestrin, "Why Should I Trust You?": Explaining the Predictions of Any Classifier' in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (KDD '16, ACM 2016) 1135.

6. Optionally, *visualise* the result as a list of feature contributions with positive (green) or negative (red) signs.

The output is a statement of the form: “For Maria’s application, the most important features driving the model’s decision were: recent credit inquiries (−0.12), credit utilisation (−0.10), income (−0.08), and credit score (+0.06).” The numbers are the coefficients of the local linear surrogate; their signs and magnitudes provide a story about which features pushed the model toward denial and which pushed against it.

Strengths and limitations

LIME’s strengths are practical: it is model-agnostic (works on any classifier you can query), local (gives an explanation for one specific prediction), and intuitive (its output is a list of weighted features that non-technical audiences readily understand). It works on tabular data, text (where perturbations are word substitutions or deletions), and images (where perturbations are random masks over patches of pixels).

Its limitations are more important than its strengths for a non-technical evaluator to internalise. First, the choice of neighbourhood — how close the perturbations are to Maria’s application — strongly affects the explanation. Second, LIME’s outputs are *unstable*: running LIME twice on the same prediction can produce different feature rankings. Third, LIME’s local linear approximation may simply be a bad fit if the model’s behaviour around Maria’s application is highly non-linear. Fourth, as Slack and colleagues showed in 2020, adversaries can construct *scaffolded models* whose actual decisions are biased but whose LIME explanations look clean.³ A bad-faith bank could, in principle, deploy such a model. The LIME explanation would attest to apparent fairness while concealing a discriminatory core.

3. SHAP — game theory comes to machine learning

If LIME was the field’s first widely-adopted technique, SHAP — SHapley Additive exPlanations, introduced by Scott Lundberg and Su-In Lee in 2017 — is its current default.⁴ SHAP has displaced LIME as the standard XAI offering in commercial ML platforms and is the technique most likely to appear in a vendor demonstration today. It draws on a much older idea: Lloyd Shapley’s 1953 cooperative game theory.

The Shapley value: a fairness puzzle from 1953

³Dylan Slack, Sophie Hilgard, Emily Jia, Sameer Singh and Himabindu Lakkaraju, ‘Fooling LIME and SHAP: Adversarial Attacks on Post Hoc Explanation Methods’ in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* (AIES 2020) 180.

⁴Scott M. Lundberg and Su-In Lee, ‘A Unified Approach to Interpreting Model Predictions’ in *Advances in Neural Information Processing Systems 30* (NeurIPS 2017) 4765.

Imagine a team of players cooperate to win a prize. Some contribute more than others. How do you fairly distribute the prize among them? Shapley's question was not trivial. Suppose a basketball team wins a championship. The leading scorer may not deserve the most credit; she may have benefited from teammates who set screens and drew defenders. The strategist who never scored may have made the others' scoring possible. How do we credit each player fairly?

Shapley proposed a unique solution.⁵ For each player, look at every possible *subset* of teammates the player could have joined, and ask: how much did the team's performance improve when this player joined this subset? Average that marginal contribution across all possible subsets. The result is the player's *Shapley value* — and Shapley proved that this number is the unique fair distribution satisfying four reasonable axioms:

- **Efficiency:** the payouts sum to the total prize.
- **Symmetry:** equivalent players get equal pay.
- **Null player:** a player who never contributes gets nothing.
- **Additivity:** the value of two combined games is the sum of the individual game values.

From championships to predictions

Substitute the basketball team with the set of input features to an ML model, and the championship prize with the model's prediction. Each feature is a “player” contributing to the final score. The Shapley value of a feature is its average marginal contribution to the prediction across all possible subsets of other features.

This produces, for each prediction, a set of per-feature contribution scores that *exactly sum* to the difference between the model's prediction and its average prediction. This is the additivity property, and it makes SHAP outputs unusually clean: any prediction can be decomposed into per-feature contributions that account for the entire deviation from the baseline.

SHAP variants and visualisation vocabulary

Computing exact Shapley values is exponentially expensive — for a model with 20 features there are roughly a million subsets to consider. SHAP introduces several approximations: TreeSHAP (exact and fast for tree-based models such as XGBoost or LightGBM, which dominate tabular ML); KernelSHAP (a model-agnostic approximation that works on any model but is slow); DeepSHAP (tailored to neural networks); and LinearSHAP (trivial for linear models).

SHAP has become the de facto visualisation standard for ML explanations because of its rich visual vocabulary. A summary (or beeswarm) plot shows feature importance across all predictions in a dataset, with each point a prediction and colour indicating the feature value — a global behaviour view. A waterfall plot shows how each feature pushes a single prediction from the

⁵Lloyd S. Shapley, 'A Value for n -Person Games' in Harold W. Kuhn and Albert W. Tucker (eds), *Contributions to the Theory of Games, Volume II* (Princeton University Press 1953) 307. Shapley shared the 2012 Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel with Alvin Roth.

baseline to the final output — the standard for explaining one decision. A force plot conveys the same information in a flat horizontal format. A dependence plot shows how the SHAP value of one feature varies with that feature's value, often coloured by a second feature to reveal interactions.

Strengths and limitations

SHAP's mathematical pedigree is its great virtue. The additivity property is enormously useful for accountability: a SHAP explanation does not merely rank features by importance; it tells you *how much* each feature contributed in absolute terms. A waterfall plot showing that “recent credit inquiries” contributed -0.12 to Maria's score is a stronger artefact than a LIME bar of comparable height.

The limitations parallel LIME's. KernelSHAP is slow for non-tree models. The “additive feature attribution” assumption may not capture complex interactions. SHAP is susceptible to the same adversarial attacks as LIME.⁶ The disagreement problem from Session 2 applies: two SHAP runs on the same prediction can produce different rankings, and SHAP can disagree with LIME on the same case. And one philosophical caveat the reader should carry into every future vendor meeting: the Shapley value's axioms are about *credit assignment in cooperative games*, not about causal explanation. Features with high SHAP values are not necessarily causal drivers of the model's prediction; they are correlated with it in a particular fairness-of-attribution sense. Mistaking SHAP attribution for causal explanation is a common error.

4. Saliency and visual explanations for vision models

The techniques so far — LIME and SHAP — present explanations as lists of feature importances. That works for tabular data with a manageable number of meaningful features. It works less well for image classifiers, where the “features” are individual pixels: a chest X-ray contains hundreds of thousands of pixels; no list of per-pixel importance scores is humanly readable. The XAI literature has developed a separate family of techniques for vision: saliency methods that highlight, on the image itself, the regions to which the model attended.

Vanilla saliency maps

The earliest approach, introduced by Karen Simonyan and colleagues in 2013, computes the gradient of the model's output with respect to each input pixel.⁷ High-gradient pixels are pixels where a small change in pixel value would change the output the most. Plot the gradient magnitudes as a heatmap overlaid on the image: a saliency map. The problem is that the gradient is noisy. Pixel-level gradients often look like static; relevant regions are present but obscured by surrounding flicker. Reading a vanilla saliency map for medical use is difficult.

⁶Lundberg and Lee (n 4).

⁷Karen Simonyan, Andrea Vedaldi and Andrew Zisserman, ‘Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps’ (2014) *Workshop Track of the International Conference on Learning Representations* (ICLR 2014), arXiv:1312.6034.

Grad-CAM: the radiology standard

Gradient-weighted Class Activation Mapping (Grad-CAM), introduced by Ramprasaath Selvaraju and colleagues in 2017, addresses the noise problem by computing gradients at a coarser level.⁸ Instead of gradients into each pixel, Grad-CAM uses gradients flowing into the last convolutional layer of a CNN — the layer just before final classification — weighted by the activation maps of that layer. The result is a low-resolution heatmap (typically 7×7 or 14×14 in the convolutional feature space) upsampled and overlaid on the image. The heatmap shows *regions* the model attended to, not individual pixels.

Grad-CAM has become the de facto standard for radiology XAI. FDA-cleared AI-assisted radiology submissions routinely include Grad-CAM heatmaps as part of their explanation interface. The clinical value is direct: a pneumonia classifier producing a heatmap concentrated on the lower right lobe agrees with a radiologist's reading of consolidation in that region; a heatmap concentrated on the metadata stamp in the corner of the image betrays a shortcut. The husky-versus-wolf failure mode of Session 2, in which a classifier learned to detect snow rather than animals, was identified using essentially this approach.

SmoothGrad and Integrated Gradients: cleaner attributions

Two further techniques refine the saliency-map approach. **SmoothGrad** (Smilkov and colleagues, 2017) reduces noise by averaging gradients over many slightly noisy copies of the input. The resulting saliency map is visibly cleaner.⁹ **Integrated Gradients** (Sundararajan and colleagues, 2017) takes a more theoretically grounded approach: it computes the integral of gradients along a path from a “baseline” image (typically a black image) to the actual input.¹⁰ This satisfies two important axioms — sensitivity (a feature affecting the output gets non-zero attribution) and implementation invariance (functionally equivalent networks give identical attributions). Integrated Gradients offers stronger theoretical guarantees than Grad-CAM, at modest computational cost.

Attention visualisation and a cautionary debate

For transformer-based models — now dominant in language and increasingly in vision — there is a tempting fourth option: visualise the *attention weights* themselves. Transformers compute attention over tokens (or image patches); plotting these weights shows which input elements the model “attended to” while producing each output element.

A long-running debate in the literature, however, urges caution. Sarthak Jain and Byron Wallace argued in a 2019 paper provocatively titled “Attention Is Not Explanation” that attention weights

⁸Ramprasaath R. Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh and Dhruv Batra, ‘Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization’ in *Proceedings of the IEEE International Conference on Computer Vision (ICCV 2017)* 618.

⁹Daniel Smilkov, Nikhil Thorat, Been Kim, Fernanda Viégas and Martin Wattenberg, ‘SmoothGrad: Removing Noise by Adding Noise’ (2017) *Workshop on Visualization for Deep Learning, ICML 2017*, arXiv:1706.03825.

¹⁰Mukund Sundararajan, Ankur Taly and Qiqi Yan, ‘Axiomatic Attribution for Deep Networks’ in *Proceedings of the 34th International Conference on Machine Learning (ICML 2017)* 3319.

and feature importances often disagree, and that high-attention tokens do not necessarily correspond to features that drive the model's output.¹¹ Sarah Wiegrefe and Yuval Pinter replied with “Attention Is Not *Not* Explanation,” refining the claim and showing that attention can carry interpretable signal under certain conditions.¹² The current consensus is that attention is one signal of model behaviour but should be interpreted cautiously, particularly in safety-critical contexts. For non-technical evaluators, the working rule is straightforward: never accept an “attention is the explanation” claim from a vendor without follow-up.

What visual explanations tell you (and what they don't)

A good visual explanation tells you *where in the image* the model looked. It does not tell you *why* that location was salient — was it the texture, the shape, the colour, the gradient? It does not tell you whether the model's reasoning was correct in any deeper sense. And it does not generalise: a heatmap is local to one image, not a statement about the model's overall behaviour. Global validation of vision models still requires the kind of careful subgroup analysis we will visit in Session 4.

5. Counterfactual explanations: the “if only” thought experiment

The fourth technique returns to one of the most familiar reasoning patterns in everyday life. “If only I had studied one more hour, I would have passed.” “If only I had left fifteen minutes earlier, I would have caught the train.” Counterfactual reasoning is how humans naturally explain causes, near-misses, and possibilities. Counterfactual explanations import the same pattern into algorithmic accountability.

Wachter, Mittelstadt and Russell (2018)

In Session 2 we encountered the work of Sandra Wachter, Brent Mittelstadt and Chris Russell on counterfactual explanations.¹³ Their 2018 paper made a striking argument: a counterfactual explanation can satisfy the legal requirement for individual explanation under GDPR (and, more recently, EU AI Act Article 86) *without* requiring the disclosure of the model's full internals. An affected person is rarely interested in understanding the model in the abstract. What they want to know is: what would have been different about my case to have changed the outcome?

“Your loan was denied. If your annual income had been £35,000 instead of £30,000, your loan would have been approved.” That single statement satisfies the moral and legal demand for explanation without exposing the bank's proprietary scoring algorithm. The counterfactual makes the decision actionable: Maria knows what she needs to do.

¹¹Sarthak Jain and Byron C. Wallace, ‘Attention Is Not Explanation’ in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics* (NAACL-HLT 2019) 3543.

¹²Sarah Wiegrefe and Yuval Pinter, ‘Attention Is Not Not Explanation’ in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing* (EMNLP 2019) 11.

¹³Sandra Wachter, Brent Mittelstadt and Chris Russell, ‘Counterfactual Explanations without Opening the Black Box: Automated Decisions and the GDPR’ (2018) 31(2) *Harvard Journal of Law and Technology* 841.

Desiderata for good counterfactuals

A counterfactual explanation is not a random nearby point on the other side of the decision boundary. To be genuinely useful, a counterfactual should be:

- **Proximate:** as close as possible to the original input. Don't suggest changing ten features when changing one is sufficient.
- **Actionable:** change features the person can actually change. "Be five years younger" or "change your country of birth" is not actionable; "reduce your credit utilisation by 15 percent" might be.
- **Plausible:** stay within realistic ranges. Don't suggest counterfactuals — an income of £10 million, or a credit utilisation of –30 percent — that lie outside the feasible space.
- **Diverse:** present multiple alternative counterfactuals so the user can choose which path to follow.

Ramaravind Mothilal, Amit Sharma and Chenhao Tan formalised the diversity requirement in their 2020 **DiCE** framework (Diverse Counterfactual Explanations).¹⁴ DiCE optimises jointly for proximity and diversity, generating not one but several counterfactuals that span the space of possible changes. The user sees alternative paths: "your loan would have been approved if your income had been £35,000; OR if your credit utilisation had been below 25 percent; OR if you had no missed payments in the past two years."

Strengths and limitations

Counterfactuals are inherently actionable, do not require model disclosure, and map cleanly to regulatory recourse requirements under GDPR Article 22 and EU AI Act Article 86. They are intuitive to affected individuals.

The limitations are real. Counterfactuals are computationally expensive, particularly when plausibility and diversity are jointly required. They may propose changes that are actionable in theory but not in practice — "increase your savings by £20,000" assumes the user has access to that capital. They do not explain why the original decision was made; they only show what would change it. And the distinction between actionable and non-actionable features is not algorithmic — it requires domain knowledge about what features a person can and cannot change in their life.

6. Choosing among the techniques

You have now met the four most-used XAI techniques. They are not interchangeable. A practical decision framework rests on three questions: who is the audience, what is the data type, and what kind of question is being asked?

¹⁴Ramaravind K. Mothilal, Amit Sharma and Chenhao Tan, 'Explaining Machine Learning Classifiers through Diverse Counterfactual Explanations' in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (FAT* 2020, ACM) 607.

Audience first

The auditor checking for bias across a population, the regulator reviewing a specific high-risk decision, the engineer debugging a misbehaving model, and the affected individual seeking recourse all need different things. A SHAP summary plot satisfies the auditor; a SHAP waterfall plot or a counterfactual satisfies the affected individual; a saliency map serves the radiologist or the model debugger.

The most common mistake in industry XAI is to produce *one explanation for all audiences*. The bank that hands a SHAP waterfall plot to a loan applicant has technically provided an explanation; in practice, the applicant cannot read it, and the explanation fails its purpose. Umang Bhatt and colleagues' industry survey of XAI deployment found exactly this pattern: explanations were most often consumed by the data scientists who produced them, not the end users they nominally served.¹⁵

Then data type

Tabular data with a manageable number of meaningful features is the home turf of SHAP. Where the model is tree-based, TreeSHAP is fast and exact; where it is not, KernelSHAP is slow but works. LIME is an alternative when SHAP is unavailable. Image data demands saliency methods: Grad-CAM as a starting point, Integrated Gradients or SmoothGrad where stronger guarantees or cleaner outputs are needed. Text data sits between: LIME and SHAP both work on text, with perturbations being word substitutions or deletions. For recourse, regardless of data type, counterfactuals are usually the right answer.

A worked example

Consider a healthcare AI system that flags chest X-rays for possible pneumonia. A clinician uses the system; one patient is flagged as high-risk and is later confirmed to have pneumonia. The hospital's safety committee wants to review the deployment. Different members of the committee need different explanations.

- The *radiologist* on the committee wants to know whether the model attended to the right regions of the X-ray. **Grad-CAM** produces a heatmap showing the model's attention concentrated on the lower right lobe — corresponding to the radiologist's reading of consolidation. The radiologist is satisfied that the model is looking in the right place.
- The *data scientist* wants to understand how the model is using clinical metadata (patient age, comorbidities, oxygen saturation) alongside the image. A **SHAP waterfall plot** shows the contributions of each metadata feature: oxygen saturation pushed the prediction up, age pushed it up further, recent antibiotic use pulled it down slightly. The data scientist is satisfied that the model is integrating metadata sensibly.

¹⁵Umang Bhatt and others, 'Explainable Machine Learning in Deployment' in *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (FAccT 2020, ACM) 648.

- The *patient's care team* wants to explain to the patient why she was flagged. A **counterfactual** is generated: "Your X-ray was flagged because of consolidation in the lower right lobe combined with low oxygen saturation; had your oxygen saturation been above 96 percent and your imaging been clear, you would not have been flagged."
- The *compliance officer* wants documentation for the next FDA audit. A **global SHAP summary** disaggregated by demographic group shows that the model performs consistently across age groups, sexes, and ethnicities. The compliance officer files this with the audit package.

The same prediction; four explanations; four audiences served. None of the four techniques is replaceable by another. A hospital that produced only the SHAP waterfall plot for all four audiences would be technically compliant but practically incomplete.

Table 1. The four techniques compared.

Technique	What it produces	Best for (data, model)	Primary audience	Key limitation
LIME	A local linear surrogate fitted around one prediction; per-feature weights	Tabular, text, image; any queryable model	Engineer debugging; quick local sanity check	Unstable across runs; sensitive to neighbourhood and distance choices
SHAP	Per-feature Shapley values that sum exactly to the prediction's deviation from baseline; global and local plots	Tabular especially; tree models with TreeSHAP	Engineer, auditor, regulator; less suitable for affected individuals	Computationally expensive on non-tree models; can be fooled adversarially
Grad-CAM (and SmoothGrad, IG)	A heatmap overlaid on the input image showing regions of model attention	Image data; convolutional or transformer vision models	Clinical experts validating; engineer debugging shortcuts	Shows where the model looked, not why; method-dependent (layer choice etc.)
Counterfactuals (incl. DiCE)	Specific alternative input(s) that would have changed the outcome	Tabular; any model where features are individually meaningful	Affected individual; legal / compliance	Says nothing about why; "actionable" requires domain knowledge

7. Validating explanations: how do you know an explanation is right?

Throughout Sessions 1 and 2 we flagged the limits of post-hoc explanation. This section gathers them into a practical checklist of caveats every consumer of XAI outputs should keep in mind.

Faithfulness

A faithful explanation reflects the underlying model's actual reasoning, to whatever degree that concept is well-defined. An unfaithful explanation looks plausible but tells a different story than what the model is doing. The disagreement problem identified by Krishna and colleagues — different XAI methods producing different explanations for the same prediction — implies that, at least sometimes, at least some explanations are unfaithful.¹⁶ Cross-validation across methods (do LIME and SHAP agree?) is one practical heuristic for catching unfaithful explanations.

Stability

A stable explanation does not change radically across reasonable variations: slightly perturbed inputs, slightly different background datasets, slightly different random seeds. SHAP and LIME both have known instability issues. Before trusting an explanation in a high-stakes context, check whether it is stable across at least these dimensions.

Adversarial robustness

Slack and colleagues' 2020 demonstration that LIME and SHAP can be deliberately fooled is not a hypothetical concern. A vendor whose business model depends on opaque decisions has incentives to deploy models whose post-hoc explanations look clean. Auditors should not rely solely on the model owner's chosen XAI tool; they should run their own explanations with different parameters and methods.

Comprehension

The audience must actually understand the explanation. As Bhatt and colleagues found, explanations are most often consumed by data scientists, not end users. User-comprehension studies — does the affected person actually understand what they have been told? — are still rare in industry deployments and should be more common.

Pitfalls and best practice

Three implementation pitfalls deserve special mention. First, background-dataset selection in SHAP matters more than most users realise: the baseline against which SHAP attributions are computed depends on what dataset you average over, and different choices produce different attributions. Second, the choice of layer in Grad-CAM matters: different convolutional layers produce different heatmaps. Document your choice and stick to it. Third, the neighbourhood size in LIME affects results substantially: smaller neighbourhoods are more local but more sensitive to noise; larger neighbourhoods are smoother but less faithful to the specific prediction.

Table 2. Common pitfalls and best practices by technique.

¹⁶Satyapriya Krishna, Tessa Han, Alex Gu, Javin Pombra, Shahin Jabbari, Steven Wu and Himabindu Lakkaraju, 'The Disagreement Problem in Explainable Machine Learning: A Practitioner's Perspective' (2022) arXiv:2202.01602.

Technique	Common pitfall	Best practice
LIME	Unstable explanations across runs; sensitive to neighbourhood and distance metric; can be fooled by scaffolded models	Run multiple seeds and report variance; document neighbourhood choice; combine with SHAP for cross-validation
SHAP	Background-dataset choice silently changes attributions; SHAP attribution mistaken for causal explanation	Document background dataset; treat SHAP values as fairness-of-credit, not causal; pair with counterfactuals for recourse
Grad-CAM	Layer choice changes heatmaps; coarse resolution may miss small but important features; risks of confirmation bias by clinicians	Pre-register layer choice; supplement with Integrated Gradients for theoretical guarantees; involve clinicians as adversarial reviewers
Counterfactuals	Counterfactuals that are mathematically valid but practically inactionable; user receives only one path even when several exist	Restrict optimisation to actionable features (with domain input); generate diverse alternatives using DiCE or equivalent

For a deeper technical treatment of each technique, Christoph Molnar's freely available textbook *Interpretable Machine Learning*, now in its third edition, remains the standard reference.¹⁷

8. Looking ahead

You now have working knowledge of the four techniques most likely to appear in your professional life. The next two sessions take this toolkit into the field. Session 4 visits sectoral case studies — healthcare, finance, manufacturing, autonomous systems, the public sector — where these techniques meet domain regulations and affected people. Session 5 closes with governance, ethics, and the integrative project.

If Session 2's central insight was the distinction between interpretability and explanation, this reading's central insight is that *explanation techniques are not interchangeable*. LIME and SHAP both produce per-feature attributions, but they answer subtly different questions. Grad-CAM shows attention; SHAP shows attribution; counterfactuals show recourse. Choosing the wrong tool produces an explanation that satisfies the producer and fails the consumer.

A useful test for the next vendor demonstration: ask three questions. Which technique is being used? Who is the intended audience for the explanation? Has the technique's faithfulness been validated? If the vendor cannot answer the first, the demonstration is a black box about a black box. If they cannot answer the second, the explanation has no audience. If they cannot answer the third, the explanation is unaudited. The toolkit is powerful. Its power, like any tool's, depends on the hands that wield it.

¹⁷Christoph Molnar, *Interpretable Machine Learning: A Guide for Making Black Box Models Explainable* (3rd edn, self-published 2025), available at <https://christophm.github.io/interpretable-ml-book/>.

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